DECK 0 OF 13 · FRC JAVA CRASH COURSE

Template gallery

Every slide layout the decks are built from

WPILib 2026 · Java · Command-based

PART 1

A section divider

Opens a part of the deck

One idea, in big letters, once or twice a deck.

The accent slide, for one key sentence.

LAYOUT: LEAD

A short paragraph with supporting points

A lead slide holds one idea in a sentence or two, for when a list would chop the reasoning up.

> A supporting point sits underneath it.

> Inline code turns blue, and **bold** works too.

LAYOUT: POINTS

A short list when the items really are parallel

- **Subsystems** own hardware.
- **Commands** say what to do with it.
- **Triggers** decide when.

LAYOUT: CARDS

Two to six cards

Subsystem

One class per mechanism. It owns the motors.

Command

One action, with a start, a loop and an end.

Trigger

Connects a button or a condition to a command.

LAYOUT: STEPS

Numbered steps, with an optional QR code

1. **Download the installer**
   Grab the 2026 WPILib release for Windows.

2. **Run it**
   Pick “Everything” so VS Code comes along.

3. **Check it**
   Open WPILib VS Code from the desktop shortcut.



WPILib install guide

docs.wpilib.org: WPILib Installation

LAYOUT: FIGURE

A docs screenshot with a numbered key

Numbers in the key match the numbers in the image.



Screenshot: WPILib documentation, CC BY 4.0.

1. **Project type:** Template
2. **Language:** Java
3. **Project base:** Command Robot

PART 2

Code layouts

LAYOUT: CODE

Full-width code

RobotContainer.java

```
private void configureBindings() {
  // Sticks read negative when pushed away, and +rotation is counter-clockwise.
  drive.setDefaultCommand(
      drive.driveArcade(
          () -> -MathUtil.applyDeadband(driver.getLeftY(), DEADBAND) * DRIVE_SCALE,
          () -> -MathUtil.applyDeadband(driver.getRightX(), DEADBAND) * TURN_SCALE));
}
```

Pulled by region from a real file in examples/.

LAYOUT: CODENOTES

Code with numbered callouts

CANDriveSubsystem.java

```
1 // CIMs are BRUSHED motors: the classic KitBot bug.
  private final SparkMax leftLeader =
      new SparkMax(LEFT_LEADER_ID, MotorType.kBrushed);
  private final SparkMax leftFollower =
      new SparkMax(LEFT_FOLLOWER_ID, MotorType.kBrushed);
  private final SparkMax rightLeader =
      new SparkMax(RIGHT_LEADER_ID, MotorType.kBrushed);
  private final SparkMax rightFollower =
      new SparkMax(RIGHT_FOLLOWER_ID, MotorType.kBrushed);

  private final DifferentialDrive drive =
2     new DifferentialDrive(leftLeader, rightLeader);
```

1. **Brushed**
   CIMs are brushed motors.

2. **Two leaders**
   Followers copy their leader.

LAYOUT: COMPARE

Side by side

Not this

```
while (!atGoal()) {
  motor.set(0.5);
}
```

This

```
run(() -> motor.set(0.5))
    .until(this::atGoal);
```

LAYOUT: TABLE

A table with a caption

Motor	Type	SPARK MAX setting
CIM	Brushed	kBrushed
NEO	Brushless	kBrushless

Rows stripe automatically.

PART 3

Recurring layouts

COMMON MISTAKE

The robot spins in circles

What you see	Push forward and the robot turns on the spot.
Why it happens	Both sides run the same direction, but the motors face opposite ways.
The fix	Invert one side in its config: .inverted(true).

BEST PRACTICE

Keep numbers in Constants

> **Every CAN ID, limit and gain gets a name in Constants.java.**

When a motor is swapped at competition you change one line, and the code explains itself.

EXERCISE

Make it drive

1. Add driveArcade to your subsystem.

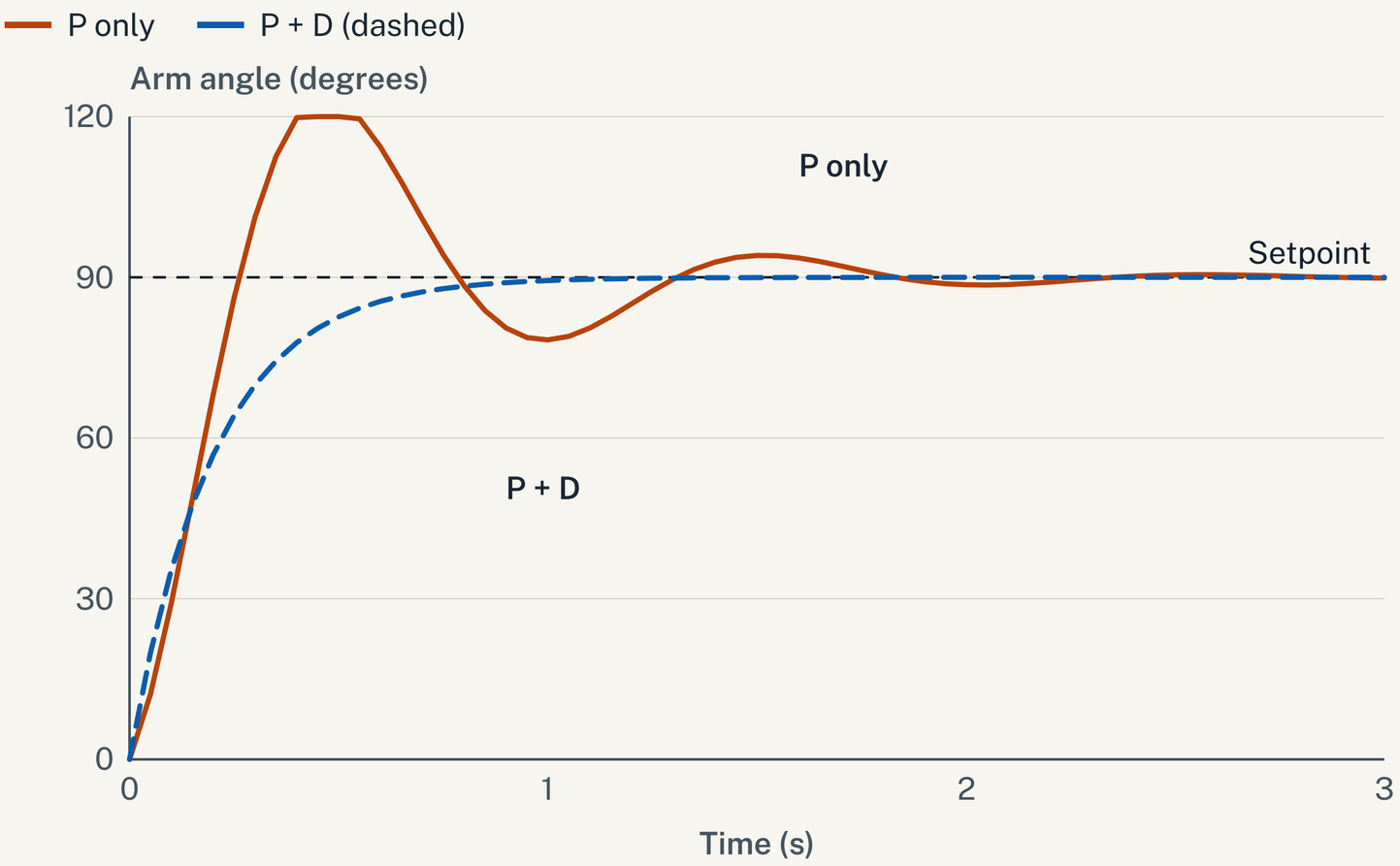
2. Set it as the default command.

3. Deploy with the wheels off the ground.

Done when: Push the stick forward and both sides spin forward.

LAYOUT: PLOT

A chart with callouts

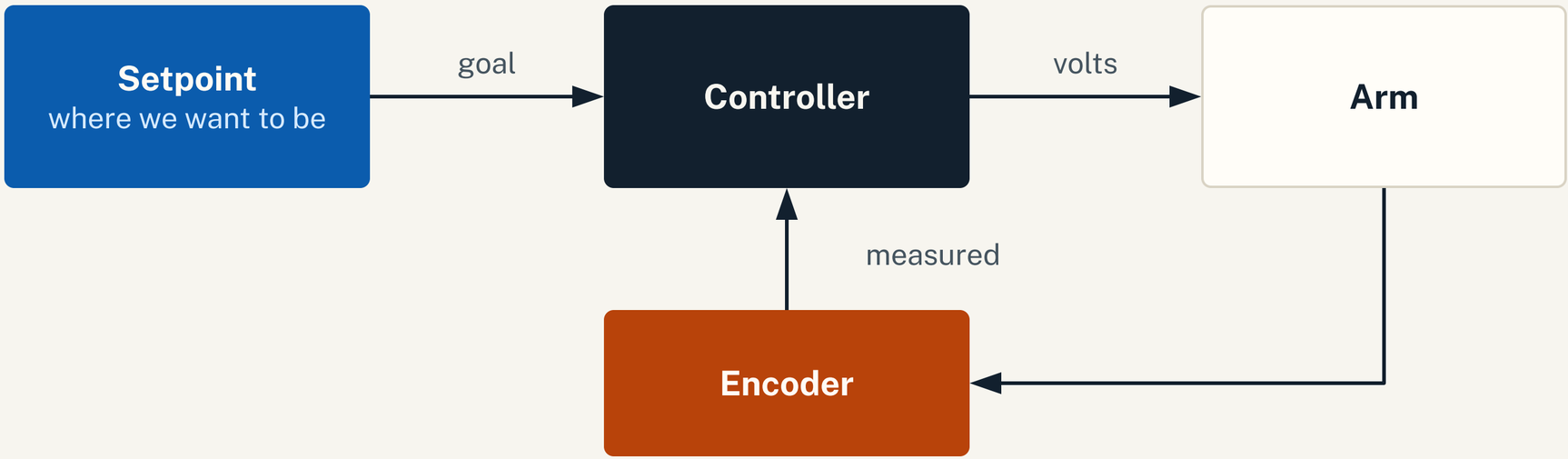


1 Overshoot
Too much P swings past the target.

2 Damping
D brakes as the error shrinks.

LAYOUT: DIAGRAM

Blocks and arrows



NEXT SEASON

What changes for 2027

In 2026	Changing for 2027
Shuffleboard	Removed: use Elastic
edu.wpi.first packages	org.wpilib packages

Everything here is still true for the 2026 roboRIO.

RECAP

What this deck covered

- Making a slide from a layout.
- Letting the build catch overflow.

Next: The real decks.

REFERENCE

Official docs



Zero to Robot

docs.wpilib.org



Command-based

docs.wpilib.org



AdvantageScope

docs.advantagescope.org